The ODD protocol (Grimm et al., 2020) presented here describes a model for simulating the foraging decision-making of primates. This ABM simulates the behavioural dynamics of foragers participating in an ecologically structured foraging task, designed to emulate a virtual reality (VR) game (the Forest Mind game). The model was developed to investigate experience-dependent foraging strategies, integrating public and personal information sources in a dynamic spatial setting. It draws upon behavioural data and design principles from initial experimental trials with human and chimpanzee participants and is intended to explore how agents adaptively balance exploration, memory, and public information to optimize resource acquisition. The model aims to investigate how agents assess the reliability of public information by learning the risks associated with it, while integrating personal memory to optimise foraging decisions.

1. Purpose and patterns

The model was developed to investigate how primates balance the use of private and public information during foraging under ecologically realistic assumptions. Public information here primarily refers to foraging knowledge that can be acquired by observing other individuals, whereas private information chiefly denotes memory of food resources. To forage, individuals navigate toward resource patches either by following a knowledgeable individual (i.e.utilizing public information), by directed movement towards a known resource patch (using private information) and/or by randomly exploring the environment. Together these different types of search giving rise to emergent behavioural patterns (Post and Semmann, 2011). Accordingly, the model was structured around two core functional components: spatial learning, which supports navigation based on environmental memory, and decision-making, which governs the dynamic allocation of reliance between public information and individual experience. Particular emphasis is placed on the modulatory role of risk perception in this trade-off, as well as the emergence of cognitive mapping. The model specifically simulates the movement dynamics and follow–explore decisions of participants in the Forest Mind game, with hypothesis testing centred on three primary behavioural metrics: foraging efficiency, the proportion of time spent following avatars, and the sequence in which participants encounter fruit-bearing trees. Here is an example path exhibited by an agent in a complete gameplay session (Fig 1.1).

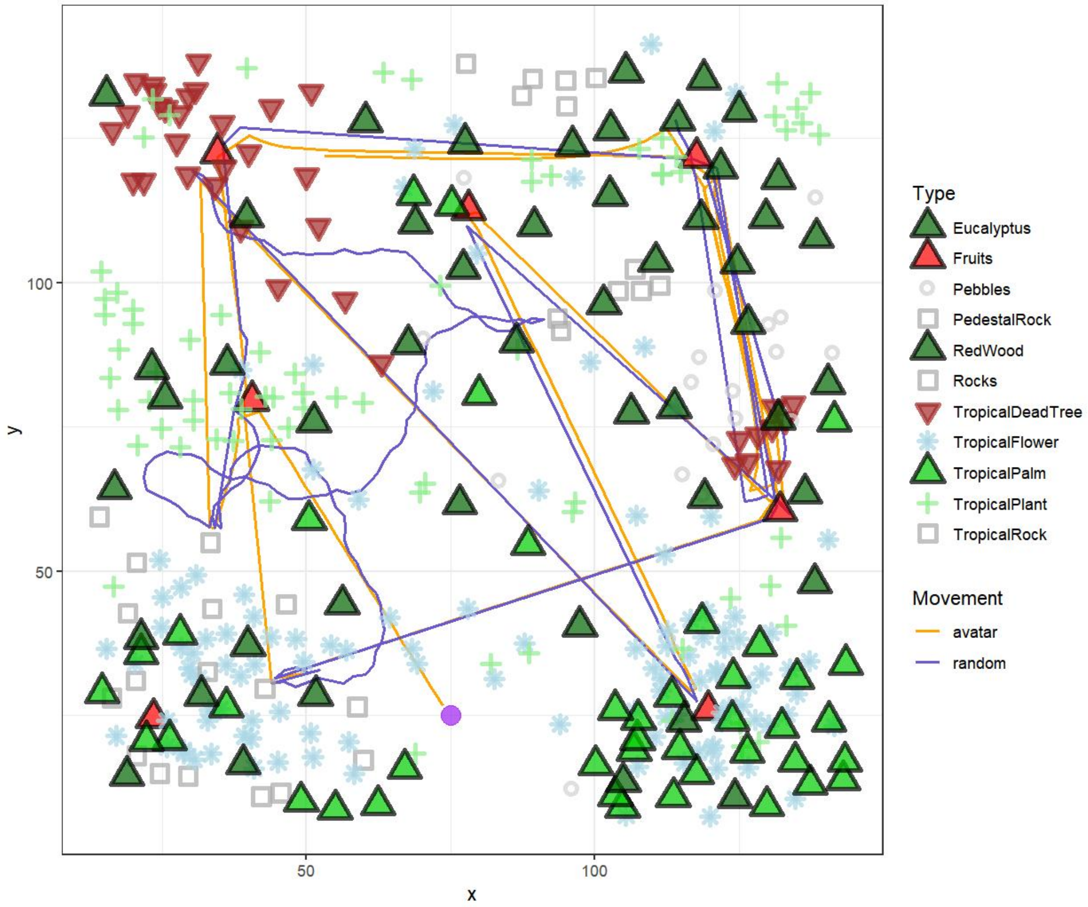


Figure 1.1. Snapshot of a Forest Mind game simulation. The purple point is the start point of simulation and the color shows different movement strategy represent by the agent.

2. Entities, state variables and scales

The model comprises three principal components: (i) agents representing foraging individuals; (ii) a knowledgeable avatar that act as carrier of public information by guiding agents toward potential resource sites; and (iii) a spatially explicit environment incorporating static foraging patches (fruit-bearing trees) and discrete environmental landmarks (like flowers, rocks and non-fruit bearing trees). Each agent maintains a dynamic internal state comprising spatial coordinates, velocity vectors, movement modality, history of tree encounters, memory structures, perceived environmental risk, and intrinsic cognitive load (Table 2.1). This state represents a cognitive map and a value estimation mechanism based on reinforcement learning. Agent decision-making is governed by a softmax rule combined with Bayesian updating, allowing agents to adaptively balance alternative foraging strategies, landmark-based navigation versus avatar-following, based on past experiences.

The knowledgeable avatar randomly chooses a fruit tree as target location (irrespective of its fruiting status) to start movement. Its state defined by their location, velocity, visibility to agents, tree encounter history, and current target. The state of fruit trees and other landmarks is defined by attributes including position, colour, size, penetrability, visibility duration, encounter frequency, eligibility traces, local co-occurrence with fruit trees (within 35 m), associative strength to neighbouring fruit trees, and associated reward value. The distance range used to determine neighbouring situation is based on the landmark distribution in the game. Overview of these properties were recorded in the Table 2.1.

The simulation is implemented on a 200 m × 200 m virtual landscape, designed to reflect empirical tree distributions observed in tropical forest environments (Fig. 2.1). This landscape includes 373 static landmarks, of which seven are fruit-bearing trees, corresponding to typical daily foraging targets reported for wild chimpanzees. Each fruit tree is surrounded by a unique constellation of co-occurring landmarks. Landmarks are taxonomically and morphologically categorised into broad classes including trees, rocks, pebbles, tropical flowers, and tropical plants. Fruit trees are visually indistinguishable from normal trees. The sole additional environmental parameter is simulation duration. Each simulation proceeds in one-second time steps for a maximum of 10 minutes. If all seven fruit trees are discovered before the allotted time, the simulation ends early and proceeds to the next iteration, reflecting the structure of the real-world game.

Table 2.1. State variables used for describing the agent, avatar, and landmarks

Symbol	Code	Description
State variables of all entities:		
g_s	sim_seed	Current gameplay session
State variables of both agent and avatar:		
x	x	x position on the grid
y	y	y position on the grid
v_x	x_vel	Velocity x component
v_y	y_vel	Velocity y component
t_v	trees_visited	ID of Trees visited by agent/avatar
State variables of agent:		
D_a	dist_to_knowledgeable	Distance to the knowledgeable avatar
s	movement_mode	movement mode chosen for a given time step. The movement mode includes two variants, random exploration and following-avatar movement.
s_m	movement_mode_mod	movement mode modifier (e.g. for "random" movement mode, "FAM" modifier)
L_c	current_landmark_id	ID of the current visited landmark
L_l	last_landmark_id	ID of the last visited landmark
D_{PI}	path_integrator_distance	Cumulative path length from the last visited landmark to the current visited landmark
$\vec{V}_{x,PI}$	path_integrator_vector_x	Total velocity of direction x from the last visited landmark to the current visited landmark
$\vec{V}_{y,PI}$	path_integrator_vector_y	Total velocity of direction y from the last visited landmark to the current visited landmark

$\vec{V}_{x,\, i \to j}$	memory_vectors_x	Memory vector of direction x from the landmark *i* to the landmark *j*
$\vec{V}_{y,\, i \to j}$	memory_vectors_y	Memory vector of direction y from the landmark *i* to the landmark *j*
h	risk	Learned risk about revisiting depleted fruit trees of following avatar
H_{cog}	intrinsic_load	Intrinsic cognitive load of agent to make decision from following avatar to self-explore
State variables of avatar:		
t_{id}	target_id	ID of selected target on the grid
x_t	target_x	x position of selected target on the grid
y_t	target_y	y position of selected target on the grid
State variables of avatar and landmarks:		
	in_dd	Whether an avatar/landmark is in agent's detection distance (0 if not, 1 if yes)
	in_fov	Whether an avatar/landmark is in agent's field of view (0 if not, 1 if yes)
	in_sight	whether an avatar/landmark is in agent's sight (in_dd and in_fov both equal to 1)
State variables of landmarks:		
pv_n	pv	Number of previous visits to the landmark *n*
p_n	is_block	Landmark n will block the move of agent or not (0 if not, 1 if yes)
c_n	color	Color vibrancy of the landmark *n* compared to the background
s_n	size	Size of the landmark *n*
T_n	time_insight	Number of iterations the landmark *n* has been in sight
S_n	sensory	Sensory input of the landmark *n* to agent
E_n	eligibility	Eligibility trace of agent to the landmark *n*
$C_{i,j}$	coccurrence	Co-occurrence of fruit tree *j* and its nearby landmark *i*
$S_{i,j}$	association	Association strength between the landmark *i* and its nearby fruit tree *j*
$R_{i,j}$	reward	Expected reward of the landmark *i* associate with its nearby fruit tree *j*

3. Process overview and scheduling

At each simulation timestep, agents move according to their currently selected strategy and consume any food resources encountered en route. Agents retain memory of the distances traversed between encountered landmarks, enabling them to construct navigational paths via path integration—a mechanism inspired by mammalian spatial cognition. In exploration mode, agent movement is additionally influenced by eligibility traces generated by sensory cues emitted from landmarks, primarily visual features such as colour and size. The strength of these eligibility traces modulates the magnitude of vector-based movement. Upon locating a fruiting tree, these memory vectors are further modulated by the agent's tendency to return to previously rewarding patches. Overall, exploratory movement is governed by a combination of memory-based vectors and vector biases toward salient patches, either previously encountered or distinguished by strong sensory cues. In the absence of established memory vectors, agent locomotion defaults to a correlated random walk (CRW) process (Turchin, 1998). During follower mode, agents navigate by aligning their movement direction with the visible avatar, following it directly. Avatars are initialized at a random position within a 1-metre radius of the agent at the beginning of each trial. They are then assigned to travel toward a randomly selected fruit tree, irrespective of fruit presence. Upon reaching the tree, the avatar disappears until the agent also encounters the same tree, at which point it reappears near the agent and repeats its guidance behaviour. Visibility of both avatars and landmarks is updated at every timestep. In addition, each landmark updates its associative memory values using the Feature Association Matrix (FAM) algorithm (Gribkova et al., 2024; see Section 7.1), which operates at each timestep. The associative reward signal derived from this process is used to compute the agent's internal cognitive load during decision-making, quantified via Shannon entropy.

A trial is defined as the path between either the starting location or the previous tree encounter and the next encountered tree. If the agent successfully reaches a fruit-bearing tree, the trial is marked as successful and one fruit is added to the foraging count. The encountered tree is then removed from the set of unseen trees and no longer treated as fruiting in subsequent trials. If no fruiting tree is found, the trial is recorded as a failure. Beyond binary success or failure, agents also evaluate whether the empty fruit tree was reached via avatar-following, allowing them to update their estimate of the risk associated with this strategy. Agents continuously adjust the value estimates of their respective movement strategies based on trial outcomes, avatar-associated risk, and the cognitive load experienced during exploratory decisions. These adjustments are implemented via reinforcement learning algorithms (see Section 7.3), allowing agents to iteratively refine their decision policies. The simulation concludes either when all seven fruit trees are discovered or when the 600-timestep limit is reached. Agents retain their learned landmark associations between simulations (represent gameplay sessions). Each simulation is repeated 10 times, corresponding to the number of game sessions played by human participants in the empirical study. One complete model run consists of these 10 repeated simulations, from which performance and behavioural data are extracted for analysis (Fig. 3.1).

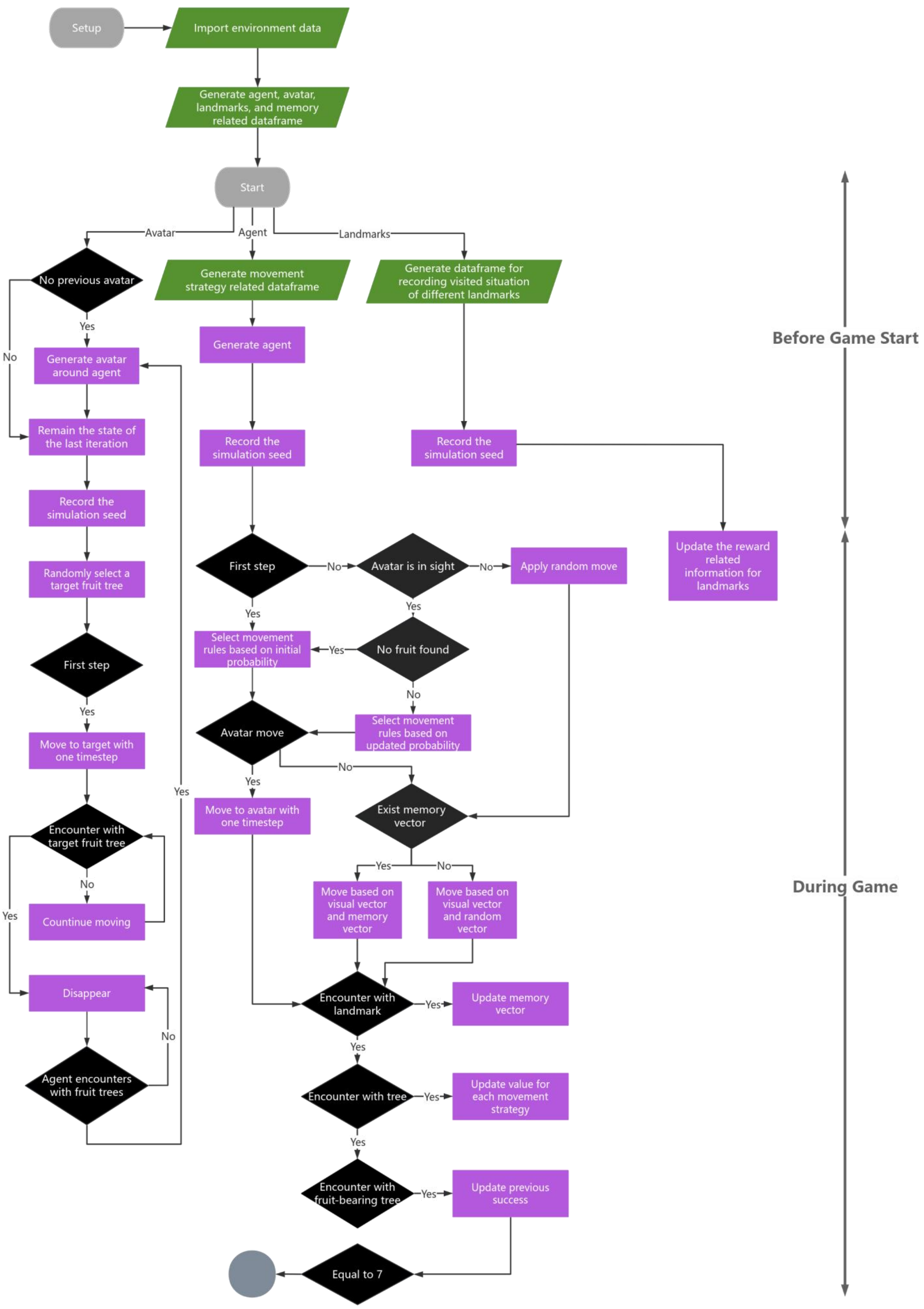


Figure 3.1. Flowchart describing the key operational processes of the agent-based model. Diamond-shaped symbols indicate logical, parallelograms indicate model input and rectangles indicate calculations. Circle indicate the end of simulation. A detailed

description of all elements of this flow diagram can be found in the following submodel descriptions.

4. Design concepts

***Basic principles*:**
This model is founded on the hypothesis that agents foraging in unfamiliar environments exhibit a tendency to follow a knowledgeable guide (avatar) when locating food resources. The avatar becomes less reliable over time (within each simulation run). As agents accumulate personal spatial knowledge over time, they begin to evaluate the relative reliability of personal versus public information and adapt their decision-making accordingly, thereby optimizing their foraging efficiency.

***Emergence*:**
Foraging efficiency emerges from two interacting processes: (i) the effective utilization of reliable public information, and (ii) accurate navigation based on spatial memory of resource locations. The utility of public information depends on the agent's capacity to assess the risk associated with relying on others, which in turn is shaped by risk sensitivity and the ability to update expectations based on failed outcomes. Navigation accuracy relies on the fidelity of path memory, itself governed by associative and reinforcement learning. Agents evaluate their memory confidence via internal uncertainty estimates. Associative learning reinforces spatial memory through repeated co-occurrence of landmarks and reward outcomes. The emergent foraging strategies reflect a balance between avatar-following and self-directed exploration.

***Adaptation*:**
Agents adaptively shift their behavioural strategies based on the perceived reliability of information sources. When public information becomes unreliable (e.g., due to avatars increasingly selecting empty trees) agents favour self-guided exploration. Conversely, when agents' memory is imprecise or uncertain, they exhibit a greater propensity to follow knowledgeable guides.

***Objectives*:**
The objective of the agent is to find all fruit-bearing trees in each successive games (7 trees within 10 games) in as short a time span as possible.

***Learning*:**
Agents incrementally learn the spatial relationships between landmarks and food rewards by constructing cognitive maps during gameplay session. Learning is both associative through repeated pairings of features and outcomes, and reflective, as agents revise their decision strategies based on trial-level successes and failures to optimize future foraging performance.

***Prediction*:**

Over the course of a game session, as fruit trees are successively encountered and avatars are more likely to guide agents to depleted (non-fruiting) trees, agents are expected to reduce reliance on avatar-following and instead prioritize decisions guided by accumulated spatial memory to maximize foraging returns. Consequently, the total number of fruit trees located should increase across successive games, and the latency to find each tree should decrease. If agents exhibit risk-averse behaviour, the proportion of avatar-following is expected to decline as more fruit trees are located. Conversely, if agents follow the Prediction Errors Induce Risk Seeking (PEIRS) model (Moeller et al., 2021), agents may persist in following avatars despite previous failures. If PEIRS model encounters positive surprises from avatar, this may trigger a disproportionate dopaminergic response (De Petrillo and Rosati, 2021). This may leads their avatar-following drop slowly or even increase near the end of a session due to high positive reward errors resulting from the discovery of fruit with avatar in later trials.

Sensing:
Agents can perceive landmarks and avatars within a limited visual range, processing their colour, size, and whether the objects are traversable. This perceptual input enables agents to decide whether to follow an avatar or engage in autonomous exploration, using visual salience to reinforce associative memories. Agents also monitor the perceived risk associated with avatar guidance and the uncertainty of their own memory, facilitating dynamic switching between movement strategies.

Interaction:
The only direct interaction between agents and avatars is the option to follow the avatar's trajectory. Both agents and avatars interact with the environment by encountering landmarks, which serve as anchors for memory and navigational learning.

Stochasticity:
Stochastic elements within the model include exploratory movement patterns, avatar selection of fruit trees, strategy switching, the number of fruit trees located, encounters with non-fruiting trees, and the learned spatial paths themselves.

Collectives:
This model does not incorporate social structures or group dynamics; agents act independently with no collective behaviour.

Observation:
The agents observe at one-second intervals and keep track of their movement and cognitive state, including movement trajectories, movement mode (exploratory or following), landmark encounters, learned risk values, memory uncertainty, and decision values associated with competing movement strategies.

5. Initialization

The model initializes a virtual environment comprising 373 spatially distributed landmarks, including both fruit-bearing and non-fruiting trees, consistent with the spatial layout and visual design of the Forest Mind game. Agent entities representing primate participants are initialized at fixed coordinates ($x = 75$, $y = 25$), corresponding to the participant starting location in the actual game. Avatars, acting as carriers of public information, are initialized at a random position within a 1 m radius from the agent's location at the start of each trial. Cognitive and learning-related parameters are set prior to model execution. These include cognitive flexibility, sensitivity to cognitive load, risk learning rate, risk sensitivity, strategy learning rate, memory decay rate, and the type of risk cognition model employed (all explained in detail in section 7 and Table 7.3.1).

6. Input data

The sole input dataset used by the model is the Forest Mind game map, comprising the coordinates and unique identifiers of all landmarks. Additional attribute columns were appended to each landmark to support agent perception and decision-making. These attributes include colour, size, passability (i.e., whether the landmark can be traversed), and classification as a tree or non-tree object. Colour values were assigned a salience score ranging from 1 to 5 based on the contrast between landmark colour and background terrain. Size values were assigned based on their relative object dimensions (Tab 6.1).

Table 6.1. Assigned visual features for the different type of landmarks

Landmarks	Type	Color	Size
Fruit Trees	Trees	1	4
Redwoods	Trees	1	4
Eucalyptus	Trees	1	4
Tropical Dead Trees	Dead Trees	3	3
Tropical Palm	Palm Trees	2	4
Rocks	Rocks	5	3
Pedestal Rocks	Rocks	5	3
Tropical Rocks	Rocks	5	3
Tropical Flowers	Flowers	4	2
Tropical Plants	Plants	2	2
Pebbles	Pebbles	2	1

7. Submodels

As previously described, the model comprises two principal components—navigation and decision-making. The architecture is subdivided into three interrelated submodels: spatial learning, movement, and strategy updating. The spatial learning and movement composed the navigation ability and the strategy updating determined the decision-making of agent. For each submodel, the objectives, operational procedures, and theoretical underpinnings are described in detail. While a comprehensive list of model parameters is provided in this section, parameters specific to particular processes are organized in corresponding subsection tables for clarity.

7.1. Spatial learning model

7.1.1. Purpose

The spatial learning module enables agents to form associative links between environmental landmarks and resource rewards, facilitating the acquisition of foraging routes. This capacity allows agents to adaptively adjust their foraging decisions based on the perceived status and features of encountered landmarks.

7.1.2. Process overview & scheduling

State variables associated with spatial learning are stored across two core data structures: a landmark dataset and a memory vector matrix. The landmark dataset encodes associative memory values linking each landmark to fruit-bearing trees, while the memory vector matrix maintains directional and distance information for navigational paths between landmarks. These spatial learning state variables are dynamically updated by the function update_landmark_and_fam_simple, which is invoked at each simulation timestep. This function operates conditionally based on the agent’s current state, executing the necessary computations to maintain and refine spatial memory representations (Fig. 7.1.2.1).

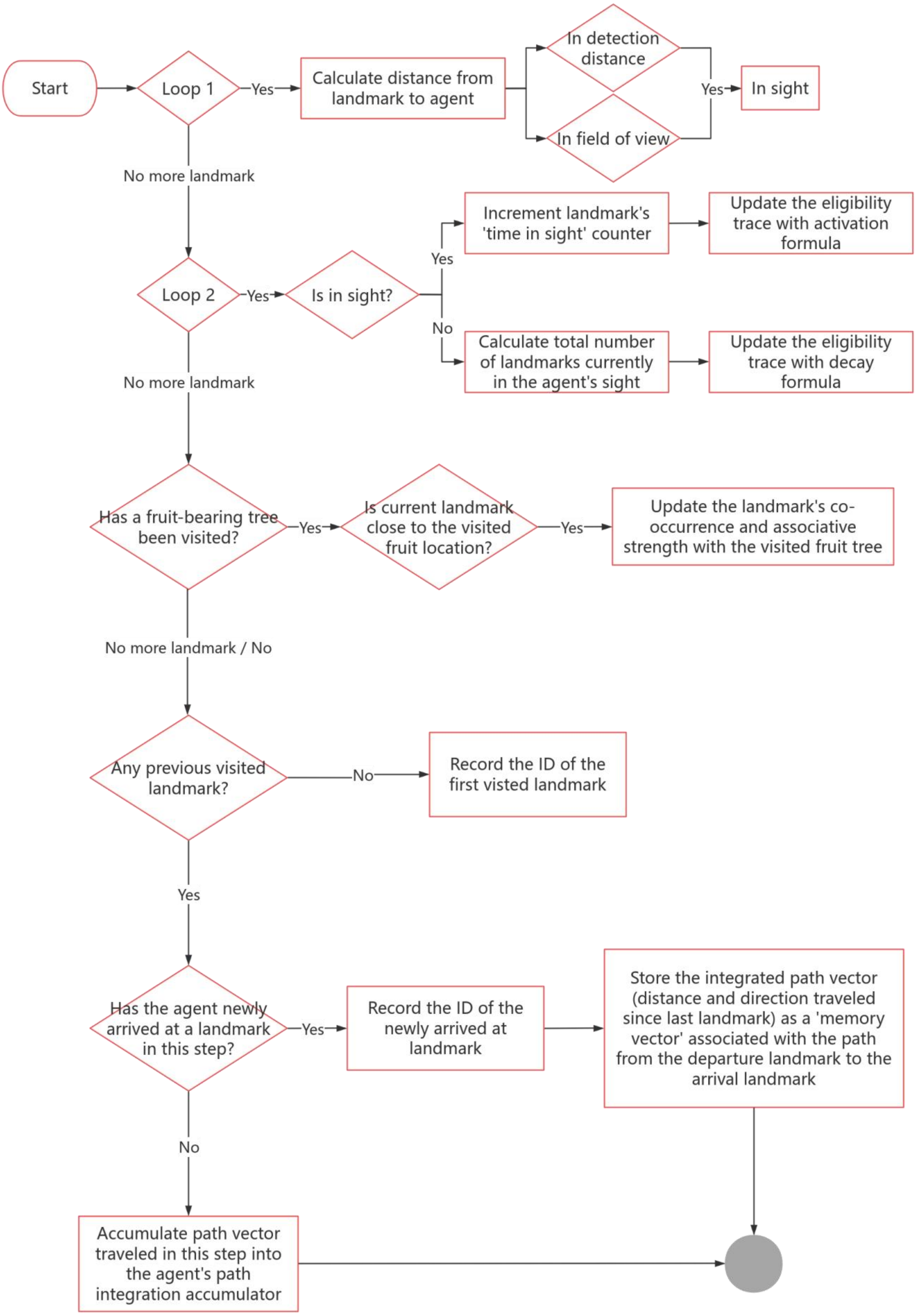


Fig 7.1.2.1 Flowchart describing the key operational processes of the spatial learning model. Diamond-shaped symbols indicate logical, rectangle indicate the calculation and the circle indicate the end of simulation. The spatial learning model is used for updating

the visual vector and memory vector mentioned in previous flow chart for general model and it updated in the beginning of each time step.

7.1.3. Associative learning

To learn the foraging map, agents integrate associative information from nearby landmarks of fruit-bearing trees. This process is mediated by a Feature Association matrix (FAM), a computationally lightweight graph-learning algorithm mimicking hippocampal auto- and hetero-associative circuits for episodic memory (Gribkova et al., 2024). Given a set of potential inputs, such as sensory cues and reward signals, the FAM encodes the temporal structure of these sequences by modulating the strength and order of associations between paired inputs. Each associative link within the matrix is further assigned an expected reward value, thereby enabling the replay of previously acquired sequences. The matrix exhibits plasticity: sequences may be subject to decay and eventual forgetting if they cease to be reinforced by repeated presentation or reward, particularly in the presence of competing input sequences. Eligibility traces for each input are derived from sensory stimuli and serve as short-term temporal records of stimulus occurrence. Although reward signals may be transient, their associated eligibility traces decay more gradually, facilitating reinforcement learning for temporally proximal cues. In the present context, sensory inputs arise from visual stimuli generated by various landmarks (landmarks and fruit trees) within the virtual environment. The sensory input strength of a given landmark *n* is computed using a model in which visual features, such as passability, conspicuous coloration, and size, are weighted to reflect their relative salience in influencing perceptual input intensity:

$$S_n = p_n \times c_n \times s_n$$

where p_n represents the landmark that can be crossed through by the agent or not;
c_n represents the conspicuity of the landmark *n* colours compared to the background;
s_n represents the size of the landmark *n*.

At each iteration, the eligibility trace E_n for a given landmark *n* is updated with following equations:

$$E_n^{t+1} = \begin{cases} k_E + \dfrac{1-k_E}{e^{a_E \cdot S_n}}, \mathit{if\ the\ landmark\ n\ is\ in\ sight} \\ E_{decay} \cdot E_n^t,\ \mathit{if\ the\ landmark\ n\ is\ not\ in\ sight} \end{cases}$$

$$E_{decay} = e^{1-N},$$

where constants k_E is the constant for scaling E_n, a_E is the constant for scaling S_n, E_{decay} is the decay variable for eligibility trace, which is higher when there's more new sensory input, N is the number of landmarks agent can see in the current iteration. The eligibility trace also used in calculating the visual vector for each landmark.

Within the FAM, each pairwise combination of landmark eligibility traces is characterised by three key parameters: Strength, Order, and Reward. These variables respectively represent: (1) the degree of co-activation or overlap between two inputs; (2) the temporal order and relative latency with which the inputs are received; and (3) the expected reward associated with their joint activation. The simplified FAM used for this model only contains the strength and reward due to the dense distribution of landmarks weakens the effect of the order in which landmarks are encountered on associative learning, as similar landmarks are seen and memorized at the same time as agent move. The associative strength between a landmark i and its surrounding fruit-bearing tree j is ultimately determined by the co-occurrence and temporal overlap of their eligibility traces E_i and E_j:

$$C_{ij}^{t+1} = C_{ij}^{t} + \Delta C_{ij}^{t},$$

$$\Delta C_{ij}^{t} = k_c(E_i \cdot E_j - E_{threshold} + k_a \cdot R_{ij})(E_i + E_j) + k_d(Strength_{ij} - C_{ij}),$$

$$Strength_{ij} = \frac{1}{1 + e^{a_s \cdot C_{ij}}},$$

where k_a,k_c,k_d,k_s denote constant parameters modulating the associative dynamics. The co-activation index C_{ij} increases in proportion to the degree of temporal overlap between the traces (i.e. $E_i + E_j > 0$), and declines when such overlap is insufficient. The associative strength between landmark i and its surrounding fruit-bearing tree j, denoted as $Strength_{ij}$, is computed as a logistic transformation of C_{ij}, bounded within the interval [0, 1]. This measure quantifies the learned associative strength between the two entities. To prevent uncontrolled escalation of associative values, the dynamics of C_{ij} are further modulated by the associative reward signal R_{ij} and $Strength_{ij}$, thereby ensuring stable convergence within the learning system. The associative reward value assigned to each landmark i is derived from the reward output and its associative strength with its neighboring fruit-bearing tree j, as follows:

$$R_{ij} = Strength_{ij} \cdot R_j,$$

Table 7.1.3.1. Parameters used in the Feature Association Matrix.

Symbol	Value	Code	Description
k_E	0.2	k_E	Constant for eligibility trace
a_E	-0.25	a_E	Exponential constant for eligibility trace
k_a	1	k_a	Constant for scaling changes in co-concurrence
k_c	0.8	k_c	Constant for reward-based changes in co-concurrence
k_d	0.001	k_d	Constant for decay of co-concurrence
a_s	-0.25	a_s	Exponential constant for Strength
$E_{threshold}$	0.04	E_threshold	Eligibility product threshold for Co-occurrence

7.1.4. Path integration and memory vectors

To enable navigation through non-overlapping spatial sequences, path integration (Etienne et al., 2004) is employed to compute the net direction and distance between pairs of landmarks and fruit trees. This module was triggered each time an agent arrived at a novel landmark and functioned as a foundation for encoding directed associations within a feature association memory. While the agent moved between landmarks (i.e., not within any arrival threshold), it continuously accumulated a path integration vector to approximate the relative displacement between successive landmarks. At each timestep t, the agent's displacement vector $\vec{v}_t = (v_t^x, v_t^y)$was added to an internal accumulator:

$$\vec{V}_{PI} \leftarrow \vec{V}_{PI} + \vec{v}_t$$

$$D_{PI} \leftarrow D_{PI} + \|\vec{v}_t\| = D_{PI} + \sqrt{(v_t^x)^2 + \left(v_t^y\right)^2}$$

where $\vec{V}_{PI}$ is the cumulative path vector and D_{PI} is the total distance travelled since the last landmark arrival. Upon confirmed arrival at a new landmark j, if a valid previous landmark i was recorded and the integrated distance D_{PI}> 0, the path integration vector $\vec{V}_{PI}$ was stored as a directional memory vector from landmark i to landmark j. These vectors encoded the relative direction between landmark pairs and served as internal guides for future memory-based navigation decisions. Reverse vectors were encoded symmetrically depending on the assumed model of bidirectional learning. Assume that the agent takes n steps, and at each step $t \in \{1, ..., n\}$, the movement vector is $\vec{v}_t = (v_t^x, v_t^y)$. Then the stored memory vector is calculated as:

$$\vec{V}_{i \to j} = \left(\sum_{t=1}^{n} v_t^x, \sum_{t=1}^{n} v_t^y \right)$$

To support learning from delayed reinforcement, a list of previously visited landmark indices was maintained in a reward history buffer. Each time the agent arrived at a distinct landmark, the index of the prior landmark was appended to this buffer. This history was subsequently used for reward backpropagation through temporally ordered landmark transitions. Upon arrival at a new landmark, the path integrator was reset to zero to ensure that memory vectors represented discrete inter-landmark transitions rather than cumulative trajectories.

7.2. Movement model

7.2.1. Purpose

The movement model enables agents to make adaptive locomotor decisions informed by spatial memory and current strategic states, while simultaneously perceiving and responding to environmental cues.

7.2.2. Process overview & scheduling

At the onset of each trial, the agent stochastically selects a movement strategy based on the current probability distribution across available strategies, random exploration and avatar-following movement. If the selected strategy is to follow the avatar, the agent further evaluates whether to maintain or switch strategies within the trial, based on its learned estimate of risk. Specifically, if the estimated risk of re-encountering a non-fruiting tree while following the avatar exceeds the expected failure probability of self-directed exploration (i.e. 1 minus the expected reward), the agent will re-sample its movement strategy according to the prevailing selection probabilities. During self-exploration, movement direction is jointly governed by visual stimulus vectors from all landmarks within perceptual range and memory vectors pointing toward the landmark associated with the highest expected reward (Fig 7.2.2.1a). In the absence of a memory vector, stochastic perturbations are introduced by generating vectorized random values drawn from a univariate truncated normal distribution. In contrast, during avatar-following, the agent moves directly towards the avatar's position (Fig 7.2.2.1b). If the agent collides with a rock, it switches to a turning-mode correlated random walk (CRW) pattern (Fig 7.2.2.1c); such movement episodes are excluded from retrospective strategy updating. At each time step, the model records the agent position, orientation, chosen strategy, current time step, and encounter status with environmental features. Encounters are defined as follows: a landmark is encountered if the agent is within 0.5 m; a tree or fruiting tree is encountered if within 5 m (reflecting the fruit-drop radius in the original game); and a rock is encountered if within 1.5 m (based on the average radius of rocks in the game environment).

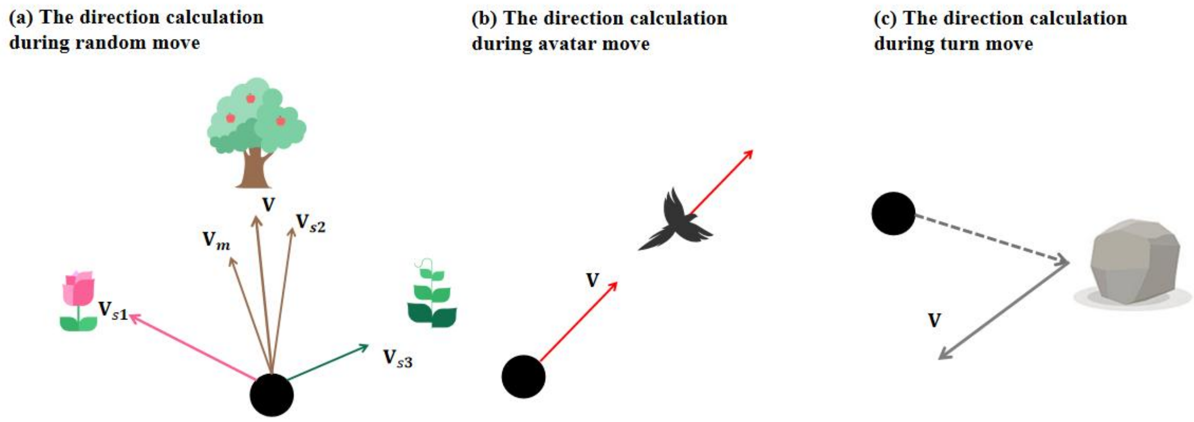


Figure 7.2.2.1. Illustration of the different movement strategy used in the agent-based model

7.2.3. The movement rule of avatar-based move and random move

Two principal types of agent movement are implemented for the agent (Fig 7.2.1): (1) goal-directed movement toward avatar, and (2) random exploratory movement, which include both memory-guided navigation and stochastic directional changes. When an agent selected a specific avatar, it moved deterministically toward that location using a normalized direction vector. The agent then updated its position and velocity based on the normalized direction vector and the step length.

Under the random move, the direction of agent was determined by both memory-based vector and visual gradient vector. To calculate the memory-based vector if the agent was located at a known landmark i, a directional vector toward another landmark j was retrieved from the feature association memory, conditioned on learned reward and associative strength. The set of available candidate transitions is denoted:

$$\mathcal{T}_i = \{j | j \neq i\},$$

For each candidate $j \in \mathcal{T}_i$, the expected value of transitioning was based on the associative reward of the landmark. The target landmark with maximal learned reward was selected:

$$j^* = \underset{j \in \mathcal{T}_i}{\text{argmax}} R_{ij}$$

The associated memory vector $\vec{m}_{i \to j^*} \in \mathrm{R}^2$ was then normalized to unit length:

$$\vec{m}_{i \to j^*} = \frac{1}{\| \vec{v} \|} \cdot \begin{bmatrix} \vec{V}_x[i, j^*] \\ \vec{V}_y[i, j^*] \end{bmatrix}, \text{where } \| \vec{v} \| = \sqrt{\vec{V}_x[i, j^*]^2 + \vec{V}_y[i, j^*]^2}$$

If no such memory vector was available, a fallback stochastic rotation (detailed below) was applied.

For visual gradient vector, The agent also computed a visual salience gradient based on currently visible landmarks. For each landmark n in view, a weighted vector was computed:

$$\vec{v}_{\mathrm{n}} = \frac{1}{d_{\mathrm{n}}^2} \cdot E_{\mathrm{n}} \cdot \hat{r}_{\mathrm{n}}, \text{where } \hat{r}_{\mathrm{n}} = \frac{(x_{\mathrm{n}} - x_t, y_{\mathrm{n}} - y_t)}{\| (x_{\mathrm{n}} - x_t, y_{\mathrm{n}} - y_t) \|}$$

Here, d_{n} is the Euclidean distance to landmark n, and $E_{\mathrm{n}} \in [0,1]$ is an eligibility trace indicating attentional salience. The overall visual vector is then:

$$\vec{v}_{\text{visual}} = \sum_k \vec{v}_k \text{ (normalized to unit length if } \| \vec{v}_{\text{visual}} \| > 0\text{)}$$

The final movement direction was computed as a weighted combination of the memory vector $\vec{v}_{\text{mem}}$ and the visual vector $\vec{v}_{\text{visual}}$:

$$\vec{v}_{\text{nav}} = (1 - w) \cdot \vec{v}_{\text{mem}} + w \cdot \vec{v}_{\text{visual}}, w = \text{visual\_weight}$$

This vector was renormalized:

$$\hat{v}_{nav} = \frac{\vec{v}_{nav}}{\| \vec{v}_{nav} \|}$$

and used to update the velocity and position of agent:

$$\vec{v}_t = \ell \cdot \hat{v}_{\text{nav}}, \ \vec{x}_{t+1} = \vec{x}_t + \vec{v}_t$$

where ℓ is the agent's fixed step length.

If both memory and visual vectors are unavailable or null (i.e., agent is uninformed because visual cues are unavailable), directional noise was introduced. A small angular perturbation $\theta \sim \mathcal{U}\left(-\frac{\phi}{2}, \frac{\phi}{2}\right)$ was applied to the agent's previous velocity vector $\vec{v}_{t-1}$:

$$\vec{v}_t = \ell \cdot \mathbf{R}_\theta \vec{v}_{t-1}, \ \mathbf{R}_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

This preserved movement continuity while enabling exploration in the absence of directed cues.

If an agent encounter a landmark that cannot be passed, such as rocks, the agent performed a directional reversal, approximating a U-turn, by sampling a heading angle $\theta \sim \mathcal{TN}(\mu = \pi, \sigma = \phi/5, [\pi - \phi/2, \pi + \phi/2])$, where $\mathcal{TN}$ is a truncated normal distribution centered at $180°$. The resulting movement vector was:

$$\vec{v}_t = \ell \cdot \mathbf{R}_\theta \vec{v}_{t-1}$$

Position remained fixed in this step $\left(\vec{x}_{t+1} = \vec{x}_t\right)$,but the heading vector was updated for the subsequent movement cycle.

Table 7.2.3.1. Parameters used in the Movement Model.

Symbol	Value	Code	Description
ℓ	2.15	step_length	Step length (in m), set up based on the velocity extracted from game
w	0.3	visual_weight	Weight for scaling the visual stimulus

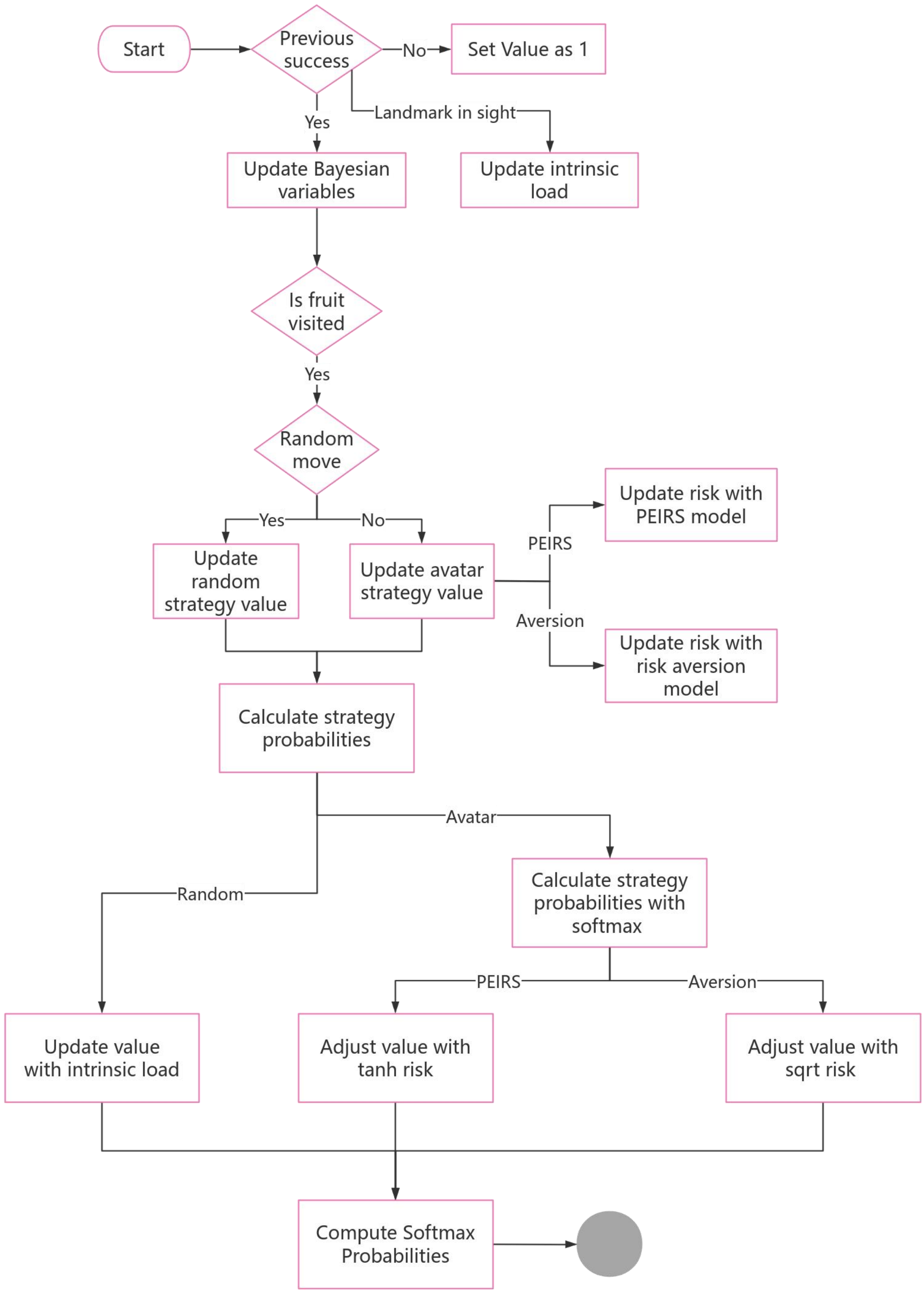
7.3. Strategy updating model

7.3.1. Purpose

The strategy update model is designed to capture the cognitive and ecological processes by which foraging agents dynamically adapt their movement decisions in response to environmental uncertainty, prior experience, and evolving estimates of risk and reward. Specifically, it aims to formalize how agents balance exploitation of known resource locations with exploration of novel areas, leveraging both individual memory and socially available cues.

7.3.2. Process overview & scheduling

The model implements an agent-based decision architecture in which individual foragers dynamically update their movement strategies through a hierarchical Bayesian reinforcement learning process. Strategy selection is influenced by spatial memory, perceived risk, and cognitive load. The process is executed iteratively at each simulation time step, consisting of the following core stages (Fig 7.3.2.1). At the beginning of each trial, the agent assesses its history of successful foraging experiences. If no prior success is recorded, strategy values are reset to uniform priors (default value = 1), enabling unbiased exploration in early learning phases. Cognitive load associated with random move are calculated based on the visual input of agent. If the detected targets were associated with reward, the intrinsic cognitive load of applying random move is computed as Shannon entropy of learned reward (see 7.3.3), representing the uncertainty of finding fruit reward through self-exploration. In the absence of discernible rewards, load is estimated using a logarithmic function of the number of visible trees, which is the maximum entropy under current situation. This load value penalizes the utility of the random movement strategy, discouraging disorganized search behaviour. If prior success exists, historical trials are partitioned into successful and failed outcomes, and the posterior belief in success of each strategy is updated via exponentially discounted parameters. The updated belief will be used for further calculation of the expected outcome of each strategy, the outcome update only happened when agent encountered fruit reward with correspond strategy. Risk was updated when the success or failure of trial was caused by avatar following behaviour. The calculation formula is based on the setup of risk update model. Intrinsic load and risk-related penalties are incorporated at the stage when all strategy values are converted to choice probabilities using a softmax transformation. Each of these steps is executed per time step as part of the agent decision-making cycle. The process ensures continuous adaptation to environmental feedback, balancing exploration and exploitation under constraints of memory, perception, and risk evaluation.



7.3.2.1. Flowchart describing the key operational processes of the spatial model. Diamond-shaped symbols indicate logical, rectangle indicate the calculation and the circle indicate the end of simulation. The updates described by this flowchart relate to the 'update value for each movement strategy' in the flowchart for general model, which runs at the end of each trial (the path between either the starting location or the previous tree encounter and the next encountered tree).

7.3.3. Strategy updating rule and softmax probability calculation

Bayesian reinforcement learning framework was adopted to simulate strategy adaptation based on prior experience and environmental uncertainty (Trimmer et al., 2011; Xiong et al., 2017). Specifically, for each movement strategy s $\in \{avatar, random\}$, the agent maintains Beta-distributed priors over success rates, updated trial-by-trial using weighted evidence. The agent updates its priors with new trial information as follows:

$$\alpha_s(t) = \lambda \cdot \alpha_s(t-1) \ + \ n_s^{success}(t),$$

$$\beta_s(t) = \lambda \cdot \beta_s(t-1) \ + \ n_s^{fail}(t),$$

where $n_s^{success}(t)$ is the number of successful decisions using strategy *s* in trial *t*, and $n_s^{fail}(t)$ is the number of failed decisions using strategy *s* in trial *t*. The decay term λ captures temporal forgetting, ensuring recent experiences influence behavior more.

The posterior belief in success for each strategy is calculated using the Beta posterior:

$$p_s(t) = \frac{\alpha_s(t)}{\alpha_s(t) + \beta_s(t)},$$

This is combined with initial prior probabilities $p_s^{(0)} \in [0, 1]$ to update the Bayesian belief, grounded in initial trust toward each strategy:

$$BayesianProb_s(t) = \frac{p_s(t) \cdot p_s^{(0)}}{p_s(t) \cdot p_s^{(0)} + (1 - p_s(t)) \cdot \left(1 - p_s^{(0)}\right)},$$

Each strategy′ s value is then updated using a Rescorla-Wagner-like rule, scaled by its Bayesian success probability:

$$V_s(t+1) = V_s(t) + k_s \cdot \big(R_s(t) - V_s(t)\big) \cdot BayesianProb_s(t),$$

where $R_s(t)$ is the trial outcome (1 for success, 0 for failure) using strategy s in trial t, ensuring agent can integrate rewards and cognitive costs to update expected results for each strategy.

Given the constraints of working memory (Sweller, 1988), learned uncertainty of environment was incorporated into the model to simulate the agent's limited capacity for memory utilization. As the primary focus of this study is the cognitive processes underlying foraging decision-making, uncertainty in environment was specifically modeled as the ambiguity imposed by working memory during decision-making. The uncertainty of decision outcomes in the random movement is primarily influenced by the availability of information within the association in this game. Conceptualizing information processing as a shift in information states allows for the application of Shannon entropy to quantify uncertainty in a computationally efficient manner (Frank, 2013; Ortega and Braun, 2013; Wilkes and Gallistel, 2017). Thus, in this study, the uncertainty in the associative rewards of certain landmark i is quantified using entropy, defined as:

$$P_i = \frac{R_i}{\sum_j^N R_j},$$

$$H = -\sum_{i=1}^{N} P_i \cdot ln P_i,$$

where N represents the number of paths can be chose from the closet landmark in sight at the iteration t;

Let Q_{random} denote the utility of self-exploration by random move at trial t, defined as a function of expected outcomes and the uncertainty associated with random move:

$$Q_{\text{random}} = V_{t,random} - b\sqrt{H_{cog}}$$

where b is the sensitivity parameter for the uncertainty in changing into the self-exploration.

The risk perception mechanism of the avatar-following movement was employed with two different model, one is the dopaminergic reward prediction frameworks described by Moeller et al. (2021). This model considered risk preferences emerge as side effects of reward prediction errors, matching with the risk-seeking behaviour caused by the hot hand effects. Another is based on the classic risk aversion model (d'Acremont et al., 2009). By integrating these elements with the softmax decision rule, the ABM captures how agents refine their foraging strategies over time, offering insights into the cognitive processes underlying movement decisions and the trade-offs between social and individual information use.

Predicted risk were updated following a Rescorla–Wagner rule (Preuschoff and Bossaerts, 2007). Let the reward r_t denote the payoff in trial t. Let the $V_{t,avatar}$ be the expected outcome of following avatar. The reward prediction error at trial t can be calculated as:

$$\delta_t = \begin{cases} r_t - V_{t,avatar}\ , if\ model\ =\ 'PEIRS' \\ r_t - R_{expect}\ , if\ model\ =\ 'Aversion' \end{cases}$$

R_{expect} was defaulted as 1 due to the fruit reward in this game continuously equal to 1. Then the risk prediction error ξ_t is denoted as the difference between the squared reward prediction error and the estimated predicted risk (d'Acremont et al., 2009):

$$\xi_t = \delta_t^2 - h_t$$

The Rescorla–Wagner rule is used to update predicted risk h_{t+1} for the next trial:

$$h_{t+1} = h_t + k_{risk}\xi_t$$

where k_{risk} is the risk learning rate.

Let Q_{avatar} denote the utility of avatar-targeted move at trial t, defined as a function of the expected outcome and the predicted risk associated with avatar-targeted move. When the agent perceive risk following the PEIRS model (Moeller et al., 2021), the utility updating rule will be:

$$Q_{avatar} = V_{t,\,avatar} + tanh(a\delta_t) \cdot h_t,$$

When the agent perceive risk following the classic risk aversion model, the utility updating rule will be:

$$Q_{avatar} = V_{t,\, avatar} - a\sqrt{h_t},$$

Where a is the risk preference parameter. In this model, $\delta = 0$ corresponds to steady state dopamine release, $\delta < 0$ means that dopamine release is suppressed, and $\delta > 0$ means that dopamine release is enhanced (Moeller et al., 2021).

The probability of selecting the specific strategy in the trial *t* was defined with the log-softmax-based probability (Banerjee et al., 2025):

$$\log p_s(t) = \beta \cdot Q_s - \log\left(\sum_{i=1}^{N} e^{\beta \cdot Q_i}\right),$$

To ensure numerical stability and prevent overflow, log-sum-exp trick was used for subtracting the maximum scaled value:

$$\log p_s(t) = \beta \cdot Q_s - \left[\max_{\mathrm{j}}\left(\beta \cdot Q_j\right) + \log\left(\sum_{i=1}^{N} e^{\beta \cdot Q_i - \max_j \beta \cdot Q_j}\right)\right],$$

Where, β is the parameter used for adjusting the cognitive flexibility of agent, higher β will let agent tend to select the decision with higher value.

Table 7.3.1. Parameters used in the Strategy Updating Model.

Symbol	Value	Code	Description
β	2	beta	Parameters regulate exploration of options when making a decision; the higher beta is, the more privileged will be the options with higher perceived value
k_{risk}	0.8	k	Learning rate of the perceived risk of following avatar
k_s	0.75	ks	Learning rate of the new experience for each strategy
a	0.9	a	Sensitivity of the learned risk
b	0.1	b	Sensitivity of the intrinsic load for changing into exploration

λ	0.8	E_threshold	Discounting factor for decay of past experiences

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